

PAST EXAMINATION QUESTIONS (FP1)**TOPIC 5: SUMMATION OF SERIES**

1. Let

$$S_N = \sum_{r=1}^N (5r+1)(5r+6) \quad \text{and} \quad T_N = \sum_{r=1}^N \frac{1}{(5r+1)(5r+6)}$$

(i) Use standard results from the List of Formulae (MF10) to show that

$$S_N = \frac{1}{3}N(25N^2 + 90N + 83). \quad [3]$$

(ii) Use the method of differences to express T_N in terms of N . $\left(\frac{1}{30} - \frac{1}{5(5N+6)}\right)$ [4](ii) Find $\lim_{N \rightarrow \infty} (N^{-3}S_N T_N)$. $\left(\frac{5}{18}\right)$ [2]**Oct/Nov 2019/11/12/13/5**

2. Let

$$u_n = \frac{4 \sin\left(n - \frac{1}{2}\right) \sin \frac{1}{2}}{\cos(2n-1) + \cos 1}.$$

(i) Using the formulae for $\cos P \pm \cos Q$ given in the List of Formulae MF10, show that

$$u_n = \frac{1}{\cos n} - \frac{1}{\cos(n-1)}. \quad [2]$$

(ii) Use the method of differences to find $\sum_{n=1}^N u_n$. $\left(-1 + \frac{1}{\cos N}\right)$ [2](iii) Explain why the infinite series $u_1 + u_2 + u_3 + \dots$ does not converge. **(It oscillates as $N \rightarrow \infty$)** [1]**May/June 2019/11/12/2**

3. (i) Use the method of differences to show that

$$\sum_{r=1}^N \frac{1}{(3r+1)(3r-2)} = \frac{1}{3} - \frac{1}{3(3N+1)}. \quad [4]$$

(ii) Find the limit, as $N \rightarrow \infty$, of

$$\sum_{r=N+1}^{N^2} \frac{N}{(3r+1)(3r-2)}. \quad [4]$$

$$\left(\frac{1}{9}\right)$$

May/June 2019/13/44. (i) By considering $(2r+1)^2 - (2r-1)^2$, use the method of differences to prove that

$$\sum_{r=1}^n r = \frac{1}{2}n(n+1). \quad [3]$$

- (ii) By considering $(2r + 1)^4 - (2r - 1)^4$, use the method of differences and the result given in part (i) to prove that

$$\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2. \quad [5]$$

The sums S and T are defined as follows:

$$S = 1^3 + 2^3 + 3^3 + 4^3 + \dots + (2N)^3 + (2N + 1)^3, \\ T = 1^3 + 3^3 + 5^3 + 7^3 + \dots + (2N - 1)^3 + (2N + 1)^3.$$

- (iii) Use the result given in part (ii) to show that $S = (2N + 1)^2(N + 1)^2$. [1]
 (iv) Hence, or otherwise, find an expression in terms of N for T , factorising your answer as far as possible. ($T = (N + 1)^2(2N^2 + 4N + 1)$) [2]
 (v) Deduce the value of $\frac{S}{T}$ as $N \rightarrow \infty$. (2) [2]

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5. Let

$$S_N = \sum_{r=1}^N (3r + 1)(3r + 4) \quad \text{and} \quad T_N = \sum_{r=1}^N \frac{1}{(3r + 1)(3r + 4)}.$$

- (i) Use standard results from the List of Formulae (MF10) to show that

$$S_N = N(3N^2 + 12N + 13). \quad [3]$$

- (ii) Use the method of differences to show that

$$T_N = \frac{1}{12} - \frac{1}{3(3N + 4)}. \quad [3]$$

- (iii) Deduce that $\frac{S_N}{T_N}$ is an integer. [2]

- (iv) Find [2]

$$\lim_{N \rightarrow \infty} \frac{S_N}{N^3 T_N}.$$

(36)

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6. Let

$$S_n = \sum_{r=1}^n (-1)^{r-1} r^2.$$

- (i) Use the standard result for $\sum_{r=1}^n r^2$ given in the List of Formulae (MF10) to show that

$$S_{2n} = -n(2n + 1). \quad [4]$$

- (ii) State the value of $\lim_{n \rightarrow \infty} \frac{S_{2n}}{n^2}$ and find $\lim_{n \rightarrow \infty} \frac{S_{2n+1}}{n^2}$. ($-2, 2$) [4]

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7. (i) Verify that

$$\frac{n(e-1)+e}{n(n+1)e^{n+1}} = \frac{1}{ne^n} - \frac{1}{(n+1)e^{n+1}}.$$

$$\left(\frac{(n+1)e-n}{n(n+1)e^{n+1}} = \frac{n(e-1)+e}{n(n+1)e^{n+1}} \right) \quad [1]$$

Let

$$S_N = \sum_{n=1}^N \frac{n(e-1)+e}{n(n+1)e^{n+1}}$$

(ii) Express S_N in terms of N and e . $\left(\frac{1}{e} - \frac{1}{(N+1)e^{N+1}} \right)$ [2]

Let $S = \lim_{N \rightarrow \infty} S_N$.

(iii) Find the least value of N such that $(N+1)(S-S_N) < 10^{-3}$. (6) [3]

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8. Find $\sum_{r=1}^n (4r-3)(4r+1)$, giving your answer in its simplest form. $\left(\frac{1}{3}n(16n^2 + 12n - 13) \right)$ [4]

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9. It is given that

$$\sum_{r=1}^n u_r = n^2(2n+3),$$

where n is a positive integer.

(i) Find [2]

$$\sum_{r=n+1}^{2n} u_r.$$

$$(14n^3 + 9n^2)$$

(ii) Find u_r . $(6r^2 - 1)$ [3]

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10. (i) Verify that

$$\frac{2r+1}{r(r+1)(r+2)} = \frac{1}{2} \left\{ \frac{(2r+1)(2r+3)}{(r+1)(r+2)} - \frac{(2r-1)(2r+1)}{r(r+1)} \right\}. \quad [2]$$

(ii) Hence show that

$$\sum_{r=1}^n \frac{2r+1}{r(r+1)(r+2)} = \frac{1}{2} \left\{ \frac{(2n+1)(2n+3)}{(n+1)(n+2)} - \frac{3}{2} \right\}. \quad [2]$$

(iii) Deduce the value of

$$\sum_{r=1}^{\infty} \frac{2r+1}{r(r+1)(r+2)}. \quad [2]$$

$\left(\frac{5}{4}\right)$
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11. Use the method of differences to find

$$\sum_{r=1}^n \frac{1}{(2r)^2 - 1}. \quad [4]$$

Deduce the value of

$$\sum_{r=1}^{\infty} \frac{1}{(2r)^2 - 1}. \quad [1]$$

$\left(\frac{1}{2}\left(1 - \frac{1}{2n+1}\right); \frac{1}{2}\right)$
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12. Express

$$\frac{4}{r(r+1)(r+2)}$$

in partial fractions and hence find

$$\sum_{r=1}^n \frac{4}{r(r+1)(r+2)}. \quad [5]$$

Deduce the value of

$$\sum_{r=1}^n \frac{4}{r(r+1)(r+2)}. \quad [1]$$

$\left(1 - \frac{2}{n+1} + \frac{2}{n+2}; 1\right)$
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13. Verify that

$$\frac{1}{(3r+1)(3r+4)} = \frac{1}{3} \left(\frac{1}{3r+1} - \frac{1}{3r+4} \right). \quad [1]$$

Let S_N denote

$$\sum_{r=1}^N \frac{1}{(3r+1)(3r+4)}$$

and let S denote

$$\sum_{r=1}^{\infty} \frac{1}{(3r+1)(3r+4)}.$$

Find the least value of N such that $S - S_N < \frac{1}{10000}$.

$$\left(S_N = \frac{1}{3} \left(\frac{1}{4} - \frac{1}{3N+4} \right); S = 12; \text{least } N = 1110\right)$$

[5]
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14. The sequence $a_1, a_2, a_3, \dots$ is such that, for all positive integers n ,

$$a_n = \frac{n+5}{\sqrt{n^2-n+1}} - \frac{n+6}{\sqrt{n^2+n+1}}.$$

The sum $\sum_{n=1}^N a_n$ is denoted by S_N . Find

- (i) The value of S_{30} correct to 3 decimal places. $\left(6 - \frac{36}{\sqrt{931}}\right)$ [3]
 (ii) The least value of N for which $S_N > 49$. (55) [4]

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15. Use the List of Formulae MF10) to show that

$$\sum_{r=1}^{13} (3r^2 - 5r + 1) \quad \text{and} \quad \sum_{r=0}^9 (r^3 - 1)$$

have the same numerical value.

[4]
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16. Use the formula for $\tan(A - B)$ in the List of Formulae (MF10) to show that

$$\tan^{-1}(x+1) - \tan^{-1}(x-1) = \tan^{-1}\left(\frac{2}{x^2}\right).$$

[3]

Deduce the sum to n terms of the series

$$\tan^{-1}\left(\frac{2}{1^2}\right) + \tan^{-1}\left(\frac{2}{2^2}\right) + \tan^{-1}\left(\frac{2}{3^2}\right) + \dots$$

[4]

$$\left(\tan^{-1}(n+1) + \tan^{-1}(n) - \frac{\pi}{4}\right)$$

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17. Given that

$$u_k = \frac{1}{\sqrt{(2k-1)}} - \frac{1}{\sqrt{2k+1}},$$

express $\sum_{k=13}^n u_k$ in terms of n . [4]

Deduce the value of $\sum_{k=13}^{\infty} u_k$. [1]

$$\left(\frac{1}{5} - \frac{1}{\sqrt{2n+1}}; \frac{1}{5}\right)$$

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18. Expand and simplify $(r + 1)^4 - r^4$. [1]

Use the method of differences together with the standard results for $\sum_{r=1}^n r$ and $\sum_{r=1}^n r^2$ to show that

$$\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2. \quad [4]$$

(4r³ + 6r² + 4r + 1)
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19. Show that the differences between the squares of consecutive integers is an odd integer. [1]

Find the sum to n terms of the series

$$\frac{3}{1^2 \times 2^2} + \frac{5}{2^2 \times 3^2} + \frac{7}{3^2 \times 4^2} + \dots + \frac{2r+1}{r^2(r+1)^2} + \dots$$

and deduce the sum to infinity of the series. [5]

$$\left(1 - \frac{1}{(n+1)^2}; 1\right)$$

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20. It is given that

$$S_n = \sum_{r=1}^n u_r = 2n^2 + n.$$

Write down the values of S_1, S_2, S_3, S_4 . Express u_r in terms of r , justifying your answer. [4]

Find

$$\sum_{r=n+1}^{2n} u_r. \quad [3]$$

$$(3, 10, 21, 36; 4r - 1; u_r = S_r - S_{r-1}; 6n^2 + n)$$

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21. Express $\frac{1}{r(r+1)(r-1)}$ in partial fractions. [1]

Find

$$\sum_{r=1}^n \frac{1}{r(r+1)(r-1)}. \quad [4]$$

State the value of

$$\sum_{r=1}^{\infty} \frac{1}{r(r+1)(r-1)}. \quad [1]$$

$$\left(\frac{1}{2(r-1)} - \frac{1}{r} + \frac{1}{2(r+1)}; \frac{1}{4} - \frac{1}{2n} + \frac{1}{2(n+1)}; \frac{1}{4}\right)$$

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22. Use the method of differences to show that

$$\sum_{r=1}^N \frac{1}{(2r+1)(2r+3)} = \frac{1}{6} - \frac{1}{2(2N+3)}. \quad [5]$$

Deduce that

$$\sum_{r=1}^{2N} \frac{1}{(2r+1)(2r+3)} < \frac{1}{8N}. \quad [4]$$

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23. Let $f(r) = r!(r-1)$. Simplify $f(r+1) - f(r)$ and hence find

$$\sum_{r=1}^{2n} r!(r^2 + 1). \quad [5]$$

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24. Let $f(r) = r(r+1)(r+2)$. Show that

$$f(r) - f(r-1) = 3r(r+1). \quad [1]$$

Hence show that

$$\sum_{r=1}^n r(r+1) = \frac{1}{3}n(n+1)(n+2). \quad [2]$$

Using the standard result for $\sum_{r=1}^n r$, deduce that

$$\sum_{r=1}^n r^2 = \frac{1}{6}n(n+1)(2n+1). \quad [2]$$

Find the sum of the series

$$1^2 + 2 \times 2^2 + 3^2 + 2 \times 4^2 + 5^2 + 2 \times 6^2 + \cdots + 2(n-1)^2 + n^2,$$

where n is odd.

[3]

$$\left(\frac{1}{2}n^2(n+1)\right)$$

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25. Show that

$$\sum_{r=n+1}^{2n} r^2 = \frac{1}{6}n(2n+1)(7n+1). \quad [4]$$

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26. Given that $f(r) = \frac{1}{(r+1)(r+2)}$, show that

$$f(r-1) - f(r) = \frac{2}{r(r+1)(r+2)}. \quad [2]$$

Hence find

$$\sum_{r=1}^n \frac{1}{r(r+1)(r+2)}. \quad [3]$$

Deduce the value of

$$\sum_{r=1}^{\infty} \frac{1}{r(r+1)(r+2)}. \quad [1]$$

$$\left(\frac{1}{2} - \frac{1}{2} \left(\frac{1}{(n+1)(n+2)} \right); \frac{1}{4}\right)$$

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27. Find the sum of the first n terms of the series

$$\frac{1}{1 \times 3} + \frac{1}{2 \times 4} + \frac{1}{3 \times 5} + \dots$$

and deduce the sum to infinity.

May/June 2012/13/1 [5]

28. Verify that

$$\frac{1}{n^2} - \frac{1}{(n+1)^2} = \frac{2n+1}{n^2(n+1)^2}. \quad [1]$$

Let

$$S_N = \sum_{r=1}^N \frac{2r+1}{r^2(r+1)^2}.$$

Express S_N in terms of N .

[2]

Let $S = \lim_{N \rightarrow \infty} S_N$. Find the least value of N such that $S - S_N < 10^{-16}$.

[3]

$$\left(1 - \frac{1}{(N+1)^2}; 10^8\right)$$

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29. Express $\frac{1}{(2r+1)(2r+3)}$ in partial fractions and hence use the method of differences to find

$$\sum_{r=1}^n \frac{1}{(2r+1)(2r+3)}. \quad [4]$$

Deduce the value of

$$\sum_{r=1}^{\infty} \frac{1}{(2r+1)(2r+3)}. \quad [1]$$

$$\left(\frac{1}{2} \left(\frac{1}{2r+1} - \frac{1}{2r+3} \right); \frac{1}{6} - \frac{1}{2(2n+3)}; \frac{1}{6} \right)$$

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30. Find $2^2 + 4^2 + \dots + (2n)^2$. [2]

Hence find $1^2 - 2^2 + 3^2 - 4^2 + \dots - (2n)^2$, simplifying your answer. [3]

$$\left(\frac{2}{3} n(n+1)(2n+1); -n(2n+1) \right)$$

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31. Use the method of differences to find S_N , where

$$S_N = \sum_{n=1}^N \frac{1}{n(n+2)}. \quad [4]$$

Deduce the value of $\lim_{N \rightarrow \infty} S_N$. [1]

$$\left(\frac{1}{2} \left(\frac{3}{2} - \frac{1}{n+1} - \frac{1}{n+2} \right); \frac{3}{4} \right)$$

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32. The sum S_N is defined by $S_N = \sum_{r=1}^N n^5$. Using the identity

$$\left(n + \frac{1}{2} \right)^6 - \left(n - \frac{1}{2} \right)^6 \equiv 6n^5 + 5n^3 + \frac{3}{8}n,$$

find S_N in terms of N . [You need not simplify your result.] [4]

$$\left(\frac{1}{6} \left\{ \left(N + \frac{1}{2} \right)^6 - \left(\frac{1}{2} \right)^6 - \frac{5N^2(N+1)^2}{4} - \frac{3}{16} N(N+1) \right\} \right)$$

Hence find $\lim_{N \rightarrow \infty} N^{-\lambda} S_N$, for each of the two cases

(i) $\lambda = 6, \left(\frac{1}{6} \right)$

(ii) $\lambda > 6, (0)$

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33. By considering the identity

$$\cos[(2n - 1)\alpha] - \cos[(2n + 1)\alpha] \equiv 2 \sin \alpha \sin 2n\alpha,$$

show that if α is not an integer multiple of π then

$$\sum_{n=1}^N \sin(2n\alpha) = \frac{1}{2} \cot \alpha - \frac{1}{2} \operatorname{cosec} \alpha \cos[(2N + 1)\alpha]. \quad [4]$$

Deduce that the infinite series

$$\sum_{n=1}^{\infty} \sin\left(\frac{2}{3}n\pi\right) \quad [1]$$

does not converge.

May/June 2010/13/2

34. Prove by induction that

$$\sum_{n=1}^N n^3 = \frac{1}{4} N^2 (N + 1)^2. \quad [5]$$

Use this result, together with the formula for $\sum_{n=1}^N n^2$, to show that

$$\sum_{n=1}^N (20n^3 + 36n^2) = N(N + 1)(N + 3)(5N + 2). \quad [3]$$

Let

$$S_N = \sum_{n=1}^N (20n^3 + 36n^2 + \mu n).$$

Find the value of the constant μ such that S_N is of the form $N^2(N + 1)(aN + b)$, where the constants a and b are to be determined. ($\mu = -12$, $a = 5$, $b = 17$) [3]

Show that, for this value of μ ,

$$5 + \frac{22}{N} < N^{-4} S_N < 5 + \frac{23}{N},$$

for all $N \geq 18$.

Oct/Nov 2009/1/11E

35. Verify that, for all positive values of n ,

$$\frac{1}{(n + 2)(2n + 3)} - \frac{1}{(n + 3)(2n + 5)} = \frac{4n + 9}{(n + 2)(n + 3)(2n + 3)(2n + 5)}.$$

For the series

$$\sum_{n=1}^N \frac{4n+9}{(n+2)(n+3)(2n+3)(2n+5)},$$

find

(i) the sum to N terms, $\left(\frac{1}{15} - \frac{1}{(N+3)(2N+5)}\right)$ [3]

(ii) the sum to infinity. $\left(\frac{1}{15}\right)$ [1]

May/June 2009/1/2

36. Given that

$$u_n = \ln\left(\frac{1+x^{n+1}}{1+x^n}\right),$$

where $x > -1$, find $\sum_{n=1}^N u_n$ in terms of N and x . [3]

$$\left(\ln\left(\frac{1+x^{N+1}}{1+x}\right)\right)$$

Find the sum to infinity of the series

$$u_1 + u_2 + u_3 + \dots$$

when

(i) $-1 < x < 1$, $(-\ln(1+x))$ [1]

(ii) $x = 1$. (0) [1]

May/June 2008/1/2

37. Express

$$\frac{2n+3}{n(n+1)}$$

in partial fractions and hence use the method of differences to find

$$\sum_{n=1}^N \frac{2n+3}{n(n+1)} \left(\frac{1}{3}\right)^{n+1}$$
 [4]

in terms of N .

Deduce the value of

$$\sum_{n=1}^{\infty} \frac{2n+3}{n(n+1)} \left(\frac{1}{3}\right)^{n+1}.$$
 [1]

$$\left(\frac{3}{n} - \frac{1}{n+1}; \frac{1}{3} \left(1 - \frac{1}{3^{N(N+1)}}\right); \frac{1}{3}\right)$$

Oct/Nov 2007/1/2

38. Verify that

$$\frac{1}{n^2+1} - \frac{1}{(n+1)^2+1} = \frac{2n+1}{(n^2+1)(n^2+2n+2)}.$$
 [1]

Use the method of differences to show that, for all $N \geq 1$,

$$\sum_{n=1}^N \frac{2n+1}{(n^2+1)(n^2+2n+2)} < \frac{1}{2}. \quad [1]$$

Write down the value of

$$\sum_{n=1}^{\infty} \frac{2n+1}{(n^2+1)(n^2+2n+2)}. \quad [1]$$

(0.5)
May/June 2007/1/1

39. Verify that if

$$v_n = n(n+1)(n+2) \dots (n+m),$$

then

$$v_{n+1} - v_n = (m+1)(n+1)(n+2) \dots (n+m). \quad [2]$$

Given now that

$$u_n = (n+1)(n+2) \dots (n+m),$$

find $\sum_{n=1}^N u_n$ in terms of m and N . [3]

$$\left(\frac{1}{m+1} (N+1)(N+2) \dots (N+m+1) - m! \right)$$

Oct/Nov 2006/1/3

40. Express

$$u_n = \frac{1}{4n^2 - 1}$$

in partial fractions, and hence find

$$\sum_{n=1}^N u_n$$

in terms of N . [4]

Deduce that the infinite series $u_1 + u_2 + u_3 + \dots$ is convergent and state the sum to infinity. [2]

$$\left(\frac{1}{2(2N-1)} - \frac{1}{2(2N+1)}; \frac{1}{2} - \frac{1}{2(2N+1)}; \frac{1}{2} \right)$$

May/June 2006/1/1

41. Show that $\left(n + \frac{1}{2}\right)^3 - \left(n - \frac{1}{2}\right)^3 \equiv 3n^2 + \frac{1}{4}$.

Use this result to prove that $\sum_{n=1}^N n^2 = \frac{1}{6}N(N+1)(2N+1)$.

The sums S, T and U are defined as follows:

$$\begin{aligned} S &= 1^2 + 2^2 + 3^2 + 4^2 + \dots + (2N)^2 + (2N+1)^2, \\ T &= 1^2 + 3^2 + 5^2 + 7^2 + \dots + (2N-1)^2 + (2N+1)^2, \\ U &= 1^2 - 2^2 + 3^2 - 4^2 + \dots - (2N)^2 + (2N+1)^2. \end{aligned}$$

Find and simplify expressions in terms of N for each of S, T and U .

$$\left(S = \frac{1}{3}(N+1)(2N+1)(4N+3); T = \frac{1}{3}(N+1)(2N+1)(2N+3); U = (N+1)(2N+1) \right)$$

Hence

- (i) describe the behaviour of $\frac{S}{T}$ as $N \rightarrow \infty$, (**converge to 2**) [1]
- (ii) prove that if $\frac{S}{U}$ is an integer then $\frac{T}{U}$ is an integer. [3]

Oct/Nov 2005/1/12E

42. Use the method of differences to find S_N , where

$$S_N = \sum_{n=N}^{N^2} \frac{1}{n(n+1)}. \quad [3]$$

Deduce the value of $\lim_{N \rightarrow \infty} S_N$. [1]

$$\left(\frac{1}{N} - \frac{1}{N^2+1}; 0 \right)$$

May/June 2005/1/1

43. Let

$$S_N = \sum_{n=1}^N (-1)^{n-1} n^3.$$

Find S_{2N} in terms of N , simplifying your answer as far as possible. ($-N^2(4N+3)$) [4]

Hence write down an expression for S_{2N+1} and find the limit, as $N \rightarrow \infty$, of $\frac{S_{2N+1}}{N^3}$. [3]

($4N^3 + 9N^2 + 6N + 1; 4$)

Oct/Nov 2004/1/5

44. Use the relevant standard results in the List of Formulae to prove that

$$S_N = \sum_{n=1}^N (8n^3 - 6n^2) = N(N+1)(2N^2-1). \quad [2]$$

Hence show that

$$\sum_{n=N+1}^{2N} (8n^3 - 6n^2)$$

can be expressed in the form

$$N(aN^3 + bN^2 + cN + d),$$

where the constants a, b, c, d are to be determined. ($a = 30, b = 14, c = -3, d = -1$)

[2]

May/June 2004/1/1

45. Given that

$$u_n = \frac{1}{n^2 - n + 1} - \frac{1}{n^2 + n + 1},$$

find $S_N = \sum_{n=N+1}^{2N} u_n$ in terms of N .

[3]

Find a number M such that $S_N < 10^{-20}$ for all $N > M$.

[3]

$$\left(\frac{1}{N^2 + N + 1} - \frac{1}{4N^2 + 2N + 1}; M + 10^6 \right)$$

Oct/Nov 2003/1/2

46. Let $v_1, v_2, v_3, \dots$ be a sequence and let

$$u_n = nv_n - (n + 1)v_{n+1},$$

for $n = 1, 2, 3, \dots$. Find $\sum_{n=1}^N u_n$.

[2]

In each of the following cases determine whether the series $u_1 + u_2 + u_3 + \dots$ is convergent, and justify your conclusion. Give the sum to infinity where this exists.

(i) $v_n = n^{-\frac{1}{2}}$. (**Diverges since $\sqrt{N+1} \rightarrow \infty$ as $N \rightarrow \infty$**)

[2]

(ii) $v_n = n^{-\frac{3}{2}}$. (**Converges to 1**)

[2]

May/June 2003/1/2

47. Given that

$$u_n = e^{nx} - e^{(n+1)x},$$

find $\sum_{n=1}^N u_n$ in terms of N and x . ($e^x - e^{(N+1)x}$)

[2]

Hence determine the set of values of x for which the infinite series

$$u_1 + u_2 + u_3 + \dots$$

[3]

is convergent and give the sum to infinity for cases where this exists.

$$(x = 0, S_{\infty} = 0, x < 1, S_{\infty} = e^x)$$

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48. Given that

$$S_N = \sum_{n=1}^N (2n-1)^3,$$

show that

$$S_N = N^2(2N^2 - 1). \quad [4]$$

Hence find

$$\sum_{n=N+1}^{2N} (2n-1)^3$$

in terms of N , simplifying your answer. $(3N^2(10N^2 - 1))$

[3]

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